

we should avoid this, since it can create issues of compliance and standardisation. The LDAP-based address book is very useful, especially when creating a centralised organisational global address list to be used with email clients such as Thunderbird, Evolution, MS Outlook, etc.

Another very important use of LDAP within organisations is to provide centralised authentication and authorisation. We're going to configure Linux clients for authentication through a centralised directory. The proposed set-up is as described below:

1. Directory Server: OpenLDAP server on RHEL 5.4 (openldap-servers-2.3.43-3.el5 rpm installed), with a configured IP address of 192.168.122.1.
2. Network client: Fedora Core 12 (Constantine) with the IP address 192.168.122.244 (leased by a DHCP server in the network).

Currently, the client uses its own authentication files: `/etc/passwd` and `/etc/shadow`. The possible entries in its user database can be checked using the `getent` command, as shown below:

```
[vbg@vbg-work openldap]$ getent passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
.
.
.
vbg:x:500:501:Varad Gupta:/home/vbg:/bin/bash
```

The `/etc/passwd` file provides certain fields required for authentication. Each line represents one database entry. As an example, look at the last line of the above output:

```
vbg:x:500:501:Varad Gupta:/home/vbg:/bin/bash
```

The format for the above entry is:

```
login (or uid):x:UID Number:GID Number:GECOS:Home Directory:Login Shell
```

The password, as we all know, comes from the `/etc/shadow` file.

To authenticate from a central LDAP directory, we need an objectClass that provides all of these required attributes. The objectClasses that provide us the attributes for Linux/UNIX authentication are: `posixAccount` and `shadowAccount`. These are defined in the `nis.schema` schema file, and are also mentioned below:

```
objectclass { 1.3.6.1.1.1.2.8 NAME 'posixAccount'
DESC 'Abstraction of an account with POSIX attributes'
```

```
SUP top AUXILIARY
MUST ( cn $ uid $ uidNumber $ gidNumber $ homeDirectory )
MAY ( userPassword $ loginShell $ gecos $ description ) )
```

```
objectclass { 1.3.6.1.1.1.2.1 NAME 'shadowAccount'
DESC 'Additional attributes for shadow passwords'
SUP top AUXILIARY
MUST uid
MAY ( userPassword $ shadowLastChange $ shadowMin $
shadowMax $ shadowWarning $ shadowInactive $
shadowExpire $ shadowFlag $ description ) )
```

We can use the above objectClasses to create a directory of users with passwords. However, Linux authentication also requires groups (as listed in the `/etc/group` file). The corresponding object class, `posixGroup` is defined thus:

```
objectclass { 1.3.6.1.1.1.2.2 NAME 'posixGroup'
DESC 'Abstraction of a group of accounts'
SUP top STRUCTURAL
MUST ( cn $ gidNumber )
MAY ( userPassword $ memberUid $ description ) )
```

As you can see from the above, `posixAccount` and `shadowAccount` objectClasses are both auxiliary. We will need a structural class to create the entry. Normally, the objectClass 'account' is used to provide a structural class to the entry.

An LDIF file containing the required entries for all users of an organisation can be created. Alternatively, various GUI tools can be used to enter this data into the directory.

Fortunately, the `openldap-servers` RPM also installs a migration tool, which contains a set of scripts in the `/usr/share/openldap/migration/` directory. These scripts are very handy when it comes to creating LDIF files from files. Let us create the required LDIF files to store authentication data centrally.

The only file that needs to be edited in the migration tool is the file `migrate_common.ph` and, that too, not extensively to begin with. The major changes that you need to specify are the following:

```
# Default DNS domain
$DEFAULT_MAIL_DOMAIN = "knafli.org";

# Default base
$DEFAULT_BASE = "dc=knafli,dc=org";
```

Since I plan to use a directory suffix of `dc=knafli,dc=org` and a DNS domain of `knafli.org`, I have set these values. In your implementation, these should be set as per the directory tree structure.